NOISE—"GROWL" NOISE AT SPEEDS OF 65-112 KM/H (40-70 MPH)—ALL WHEEL DRIVE (AWD) EQUIPPED VEHICLES ONLY	Article No. 02-10-1

MERCURY: 2002 MOUNTAINEER

Article **02-1-3** is being republished in its entirety to correct the tool numbers used.

ISSUE

Some AWD vehicles may exhibit a "growl" noise at speeds of 65-112 km/h (40-70 mph). This may be caused by the design of the Transfer Case Drive Chain and the hardness of the Front Axle Pinion Bushing.

ACTION

Verify the condition. If normal diagnostics cannot pinpoint the condition, replace the Front Axle Pinion Bushing, Transfer Case Drive Chain and perform Powertrain Neutralization Procedures. Refer to the following Service Procedure for details.

SERVICE PROCEDURE

Replace Transfer Case Drive Chain: Part Number 1L2Z-7A029-LA

2002 Mountaineer Transfer Case Removal

1. Remove the rear driveshaft. For additional information, refer to Workshop Manual Section 205-01.
2. Remove the skid plate, if equipped.
3. Index-mark the front axle pinion flange and the front driveshaft.
4. Index-mark the front output shaft assembly and the front driveshaft constant velocity (CV) joint.
5. Remove and discard the bolts and washers.
6. Remove and discard the bolts and universal joint retainers.

CAUTION
ALWAYS DISCONNECT THE FRONT DRIVESHAFT FROM THE TRANSFER CASE FIRST. OTHERWISE, THE WEIGHT OF THE DRIVESHAFT CAN PINCH THE BOOT BETWEEN THE SHAFT AND THE HOUSING AND CAN CAUSE THE BOOT TO TEAR.

CAUTION
TAPE THE BEARING CUPS TO THE DRIVESHAFT TO PREVENT THEM FROM FALLING OFF OF THE SPIDER.

NOTE
THE FRONT DRIVESHAFT CV JOINT IS NOT REPAIRABLE.

7. Remove the front driveshaft.
8. Drain the fluid if the transfer case is to be disassembled.
9. Disconnect the motor electrical connector, if equipped.
10. Disconnect the vent hose.
11. Position a suitable jack under the transmission.
12. Remove the LH heat shield-to-crossmember bolts.
13. Remove the RH heat shield-to-crossmember bolts.
14. Remove the two nuts and the RH crossmember cover, then remove the four crossmember bolts.
15. Remove the four LH crossmember bolts.
16. Remove the transmission mount nuts, then remove the crossmember. It may be necessary to pry the crossmember from the vehicle.
17. Remove the transmission mount.
18. Position a suitable transmission jack to the transfer case. Securely strap the transfer case to the jack.
19. Lower the transmission jacks down just enough to access the upper transfer case -to-transmission bolts, including the bolt retaining the fuel line bracket. Position the bracket out of the way.
20. Remove the remaining transfer case -to-transmission bolts. Separate the transfer case from the transmission.

Copyright © 2002 Ford Motor Company

21. Remove the transfer case from the vehicle. Move the transfer case rearward until the transfer case clears the output shaft. Lower and at the same time slightly tilt the rear of the transfer case downward.
22. Clean the transfer case and transmission mating surfaces.

Refer to Transfer Case "DISASSEMBLY AND ASSEMBLY" Section 308-07C to replace Transfer Case Chain.

2002 Mountaineer Transfer Case Installation

1. Position the transfer case into the vehicle and install the fuel line bracket and the transfer case -to- transmission bolts. Torque bolts to 41 N•m (30 Lb-ft).
2. Using suitable jacks, raise the transfer case and transmission completely into position.
3. Remove the transmission jack from under the transfer case.
4. Install the transmission mount. Torque the mount-to-case bolts to 90 N•m (66 Lb-ft).
5. Position the crossmember and loosely install the transmission mount nuts.
6. Install the eight crossmember bolts. Torque to 63 N•m (46 Lb-ft).
7. Tighten the transmission mount nuts. Torque the mount -to- crossmember bolts to 98 N•m (72 Lb-ft).
8. Remove the transmission jack from under the transmission.
9. Install the RH crossmember cover and two nuts. Torque to 9 N•m (80 Lb-in).
10. Install the RH heat shield -to- crossmember bolts. Torque to 20 N•m (15 Lb-ft).
11. Install the LH heat shield -to- crossmember bolt. Torque to 20 N•m (15 Lb-ft).
12. Connect the vent hose.
13. Connect the shift motor electrical connector, if equipped.

NOTE
DO NOT INSTALL THE FRONT DRIVESHAFT AT THIS TIME. CONTINUE WITH PROCEDURE.

CAUTION
TRANSFER CASE FAILURE CAN RESULT IF THE CORRECT FILL PROCEDURES ARE NOT FOLLOWED.

NOTE
THE FLUID LEVEL MUST BE JUST BELOW THE FILL PLUG.

14. Remove the fill plug and check the fluid level. If the fluid is below the proper level, fill the transfer case with transmission fluid.
15. Install the rear driveshaft. For additional information, refer to Workshop Manual Section 205-01.

Replace Front Axle Pinion Bushing: Part Number 1L2Z-3A443-BA

1. Raise vehicle on a hoist suitable for access to the front drive axle assembly.
2. Remove the front frame crossmember (4 bolts).
3. Disconnect front suspension stabilizer bar at the frame clamps (4 bolts) and link bolts (2 bolts), drop bar and allow it to rest on the lower suspension "A" arms.
4. Support axle with jack stand or transmission jack.

NOTE
DAMAGE TO CV JOINT BOOTS CAN OCCUR IF AXLE ASSEMBLY IS ALLOWED TO REST ON THE BOOTS.

WARNING
SECURE THE AXLE TO THE JACK WITH A SAFETY STRAP. FAILURE TO FOLLOW THESE INSTRUCTIONS MAY RESULT IN PERSONAL INJURY.

5. Remove the three axle attachment bolts at the isolation bushings, allow the axle to drop out of the frame brackets, and roll the pinion nose down for access to the Pinion Bushing. (Figure 2).

CAUTION
ANYTIME BOLTS, WASHERS, SPACERS, OR NUTS ARE LOOSENED IN THE FRONT DIFFERENTIAL SUPPORT FOR ANY REASON, INSTALL NEW COMPONENTS TO PREVENT DAMAGE.

6. Note the indexing and flange depth of the current Pinion Bushing and remove it using the following tools (Figure 3):
   - 205-376
   - 205-371
   - 205-374
7. Install the new Pinion Bushing using the same rotational indexing and depth as the original bushing with the following tools (Figure 4):
   - 204-187
   - 205-369
   - 205-485/6
8. Using all new hardware, reinstall the axle, stabilizer bar, and the crossmember by reversing Steps 1 through 5. The stabilizer link bolts must be tightened while the vehicle is resting on the tires with the suspension in the normal curb height condition.

CAUTION
THE CAN (DOMED CV JOINT HOUSING COVER) ON THE FRONT DRIVESHAFT IS PRESSED INTO THE CV JOINT HOUSING AT THE FACTORY. WHEN HOUSED CORRECTLY, THE CAN WILL APPEAR AS SHOWN IN FIGURE 1, "TOP BOX". DO NOT RESEAT THE CAN IN THE CV JOINT HOUSING IF THE CAN'S FLANGE IS ABOVE THE CV JOINT HOUSINGS SHOWN IN FIGURE 1, "BOTTOM BOX", INSTALL A NEW DRIVESHAFT.

9. Verify that the can flange is housed correctly in the CV joint housing as shown in Figure 1, top box. Install a new driveshaft if the can flange is not housed correctly.

CAUTION
ALWAYS CONNECT THE FRONT DRIVESHAFT TO THE AXLE FIRST. OTHERWISE THE WEIGHT OF THE DRIVESHAFT CAN PINCH THE BOOT BETWEEN THE SHAFT AND HOUSING AND CAN CAUSE THE BOOT TO TEAR.

NOTE
ALIGN THE INDEX MARKS.

10. Install the front driveshaft.

CAUTION
TIGHTEN THE BOLTS EVENLY IN A CROSS PATTERN OR DAMAGE WILL OCCUR TO THE CV JOINT.

NOTE
INSTALL NEW WASHERS AND BOLTS. IF NEW BOLTS ARE NOT AVAILABLE, COAT THE THREADS OF THE ORIGINAL BOLTS WITH THREADLOCK AND SEALER.

11. Install the new washers and bolts. Tighten the bolts evenly in a cross pattern.
12. Verify that the CV joint has seated squarely in the transfer case flange.
    a. Check that the space between the end of the CV joint cap and the end of the transfer case flange is equal from the top to the bottom of the two components.
    b. Rotate the driveshaft 1/4 turn.
    c. Check that the space between the end of the CV joint cap and the end of the transfer case flange is equal from the top of the bottom of the two components.
    d. Repeat this procedure several times.
    e. If the space is not equal at any point in the attachment, remove and reinstall the driveshaft as described in this procedure.

NOTE
INSTALL NEW UNIVERSAL JOINT RETAINERS AND BOLTS. IF NEW BOLTS ARE NOT AVAILABLE, COAT THE THREADS OF THE ORIGINAL BOLTS WITH THREADLOCK AND SEALER.

13. Install the new universal joint retainer and bolts.
14. Install the skid plate, if equipped.

Perform Powertrain Neutralization Procedure

Refer to section 100-04 of the Workshop Manual for Powertrain Neutralization procedures. Verify that the Service Procedure has corrected the condition.

PART NUMBER	PART NAME
1L2Z-7A029-LA	Transfer Case Drive Chain
1L2Z-3A443-BA	Front Axle Pinion Bushing

OTHER APPLICABLE ARTICLES: NONE
SUPERSEDES: 02-1-3
WARRANTY STATUS: Eligible Under The Provisions Of Bumper To Bumper Warranty Coverage

OPERATION	DESCRIPTION	TIME
021001A	Replace The Front Drive Axle Pinion Bushing, Transfer Case Drive Chain And Perform Powertrain Neutralization Procedure	4.5 Hrs

DEALER CODING

BASIC PART NO.	CONDITION CODE
3A433	42

OASIS CODES: 508000, 509000, 597997, 702000, 702100, 703000, 703300

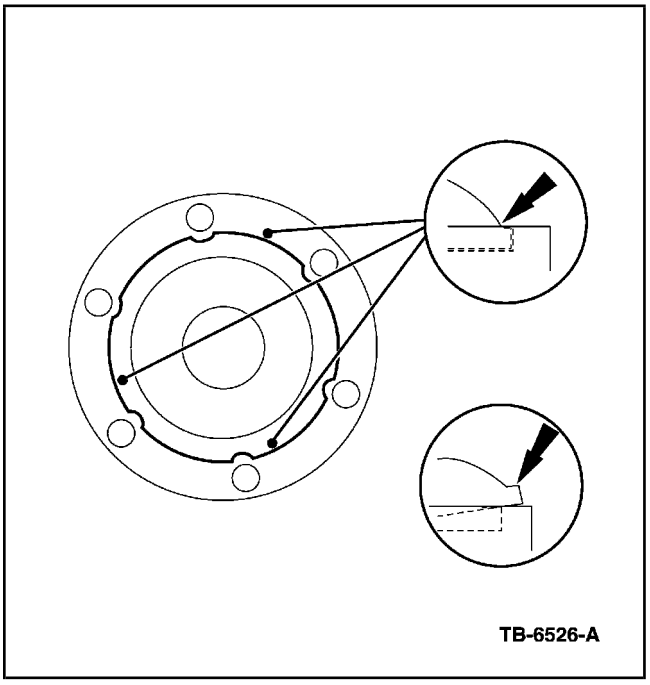


Figure 1 - Article 02-10-1

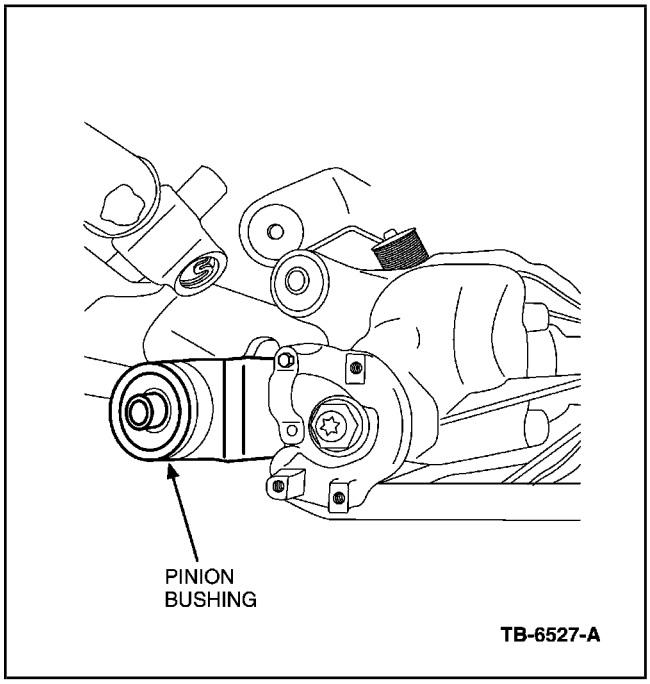


Figure 2 - Article 02-10-1

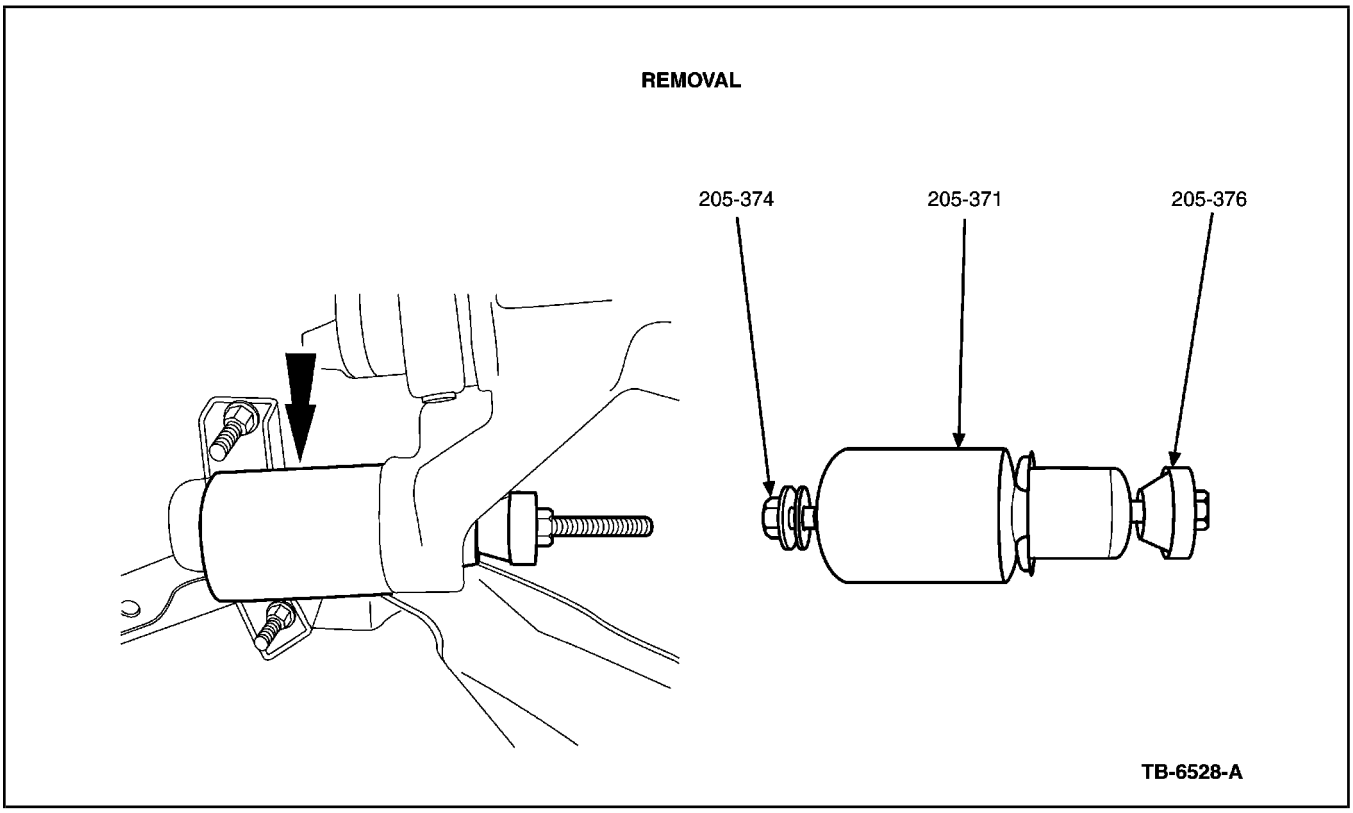


Figure 3 - Article 02-10-1

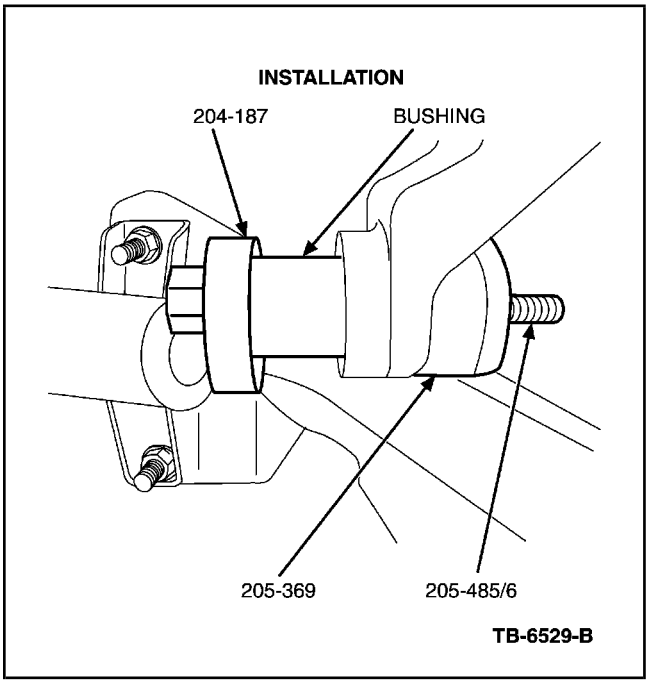


Figure 4 - Article 02-10-1